Architecture

NetApp Solutions

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Architecture

Overview

Part of trusting a cloud solution is understanding the architecture and how it is secured. This section calls out different aspects of the Cloud Volumes Service architecture in Google to help alleviate potential concerns about how data is secured, as well as call out areas where additional configuration steps might be required to obtain the most secure deployment.

The general architecture of Cloud Volumes Service can be broken down into two main components: the control plane and the data plane.

Control plane

The control plane in Cloud Volumes Service is the backend infrastructure managed by Cloud Volumes Service administrators and NetApp native automation software. This plane is completely transparent to end users and includes networking, storage hardware, software updates, and so on to help deliver value to a cloud-resident solution such as Cloud Volumes Service.

Data plane

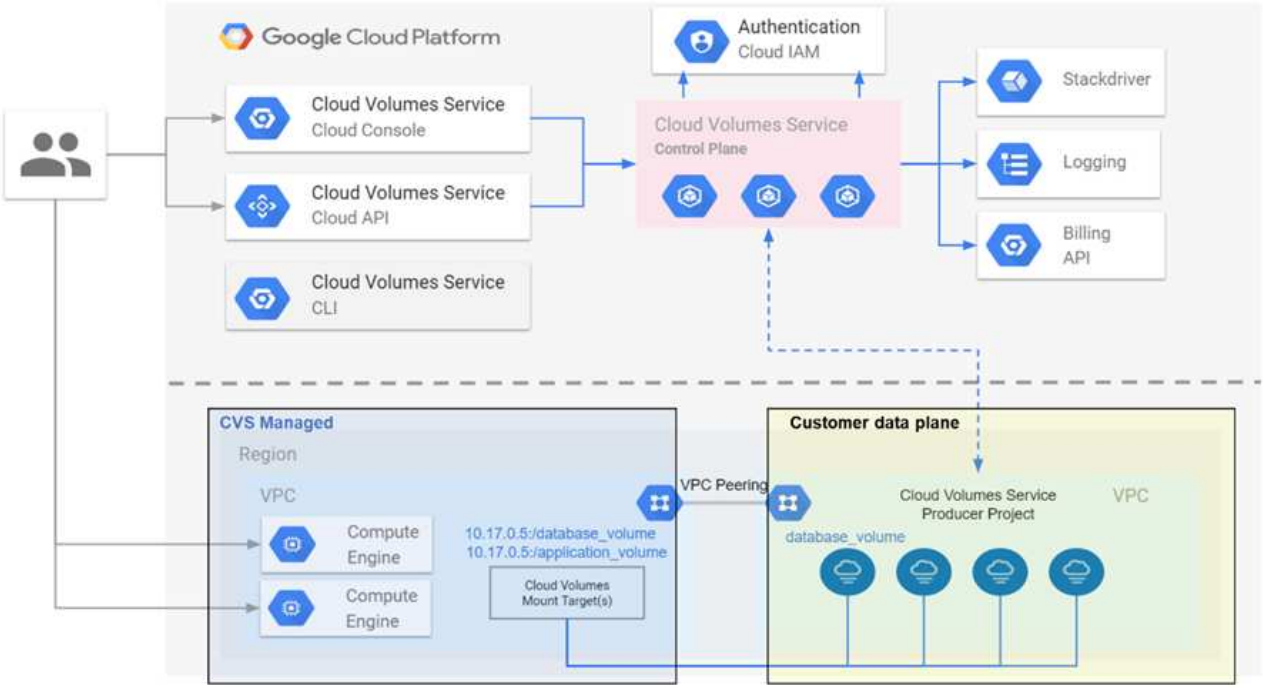
The data plane in Cloud Volumes Service includes the actual data volumes and the overall Cloud Volumes Service configuration (such as access control, Kerberos authentication, and so on). The data plane is entirely under the control of the end users and the consumers of the Cloud Volumes Service platform.

There are distinct differences in how each plane is secured and managed. The following sections cover these differences, starting with a Cloud Volumes Service architecture overview.

Cloud Volumes Service architecture

In a manner similar to other Google Cloud native services such as CloudSQL, Google Cloud VMware Engine (GCVE), and FileStore, Cloud Volumes Service uses Google PSA to deliver the service. In PSA, services are built inside a service producer project, which uses VPC network peering to connect to the service consumer. The service producer is provided and operated by NetApp, and the service consumer is a VPC in a customer project, hosting the clients that want to access Cloud Volumes Service file shares.

The following figure, referenced from the architecture section of the Cloud Volumes Service documentation, shows a high-level view.



The part above the dotted line shows the control plane of the service, which controls the volume lifecycle. The part below the dotted line shows the data plane. The left blue box depicts the user VPC (service consumer), the right blue box is the service producer provided by NetApp. Both are connected through VPC peering.

Tenancy model

In Cloud Volumes Service, individual projects are considered unique tenants. This means that manipulation of volumes, Snapshot copies, and so on are performed on a per- project basis. In other words, all volumes are owned by the project that they were created in and only that project can manage and access the data inside of them by default. This is considered the control plane view of the service.

Shared VPCs

On the data plane view, Cloud Volumes Service can connect to a shared VPC. You can create volumes in the hosting project or in one of the service projects connected to the shared VPC. All projects (host or service) connected to that shared VPC are able to reach the volumes at the network layer (TCP/IP). Because all clients with network connectivity on the shared- VPC can potentially access the data through NAS protocols, access control on the individual volume (such as user/group access control lists (ACLs) and hostnames/IP addresses for NFS exports) must be used to control who can access the data.

You can connect Cloud Volumes Service to up to five VPCs per customer project. On the control plane, the project enables you to manage all created volumes, no matter which VPC they are connected to. On the data plane, VPCs are isolated from one another, and each volume can only be connected to one VPC.

Access to the individual volumes is controlled by protocol specific (NFS/SMB) access control mechanisms.

In other words, on the network layer, all project s connected to the shared VPC are able to see the volume, while, on the management side, the control plane only allows the owner project to see the volume.

VPC Service Controls

VPC Service Controls establish an access control perimeter around Google Cloud services that are attached to the internet and are accessible worldwide. These services provide access control through user identities but cannot restrict which network location requests originate from. VPC Service Controls close that gap by introducing the capabilities to restrict access to defined networks.

The Cloud Volumes Service data plane is not connected to the external internet but to private VPCs with well-defined network boundaries (perimeters). Within that network, each volume uses protocol-specific access control. Any external network connectivity is explicitly created by Google Cloud project administrators. The control plane, however, does not provide the same protections as the data plane and can be accessed by anyone from anywhere with valid credentials (JWT tokens).

In short, the Cloud Volumes Service data plane provides the capability of network access control, without the requirement to support VPC Service Controls and does not explicitly use VPC Service Controls.

Packet sniffing/trace considerations

Packet captures can be useful for troubleshooting network issues or other problems (such as NAS permissions, LDAP connectivity, and so on), but can also be used maliciously to gain information about network IP addresses, MAC addresses, user and group names, and what level of security is being used on endpoints. Because of the way Google Cloud networking, VPCs, and firewall rules are configured, unwanted access to network packets should be difficult to obtain without user login credentials or JWT tokens into the cloud instances. Packet captures are only possible on endpoints (such as virtual machines (VMs)) and only possible on endpoints internal to the VPC unless a shared VPC and/or external network tunnel/IP forwarding is in use to explicitly allow external traffic to endpoints. There is no way to sniff traffic outside of the clients.

When shared VPCs are used, in-flight encryption with NFS Kerberos and/or SMB encryption can mask much of the information gleaned from traces. However, some traffic is still sent in plaintext, such as DNS and LDAP queries. The following figure shows a packet capture from a plaintext LDAP query originating from Cloud Volumes Service and the potential identifying information that is exposed. LDAP queries in Cloud Volumes Service currently do not support encryption or LDAP over SSL. CVS-Performance support LDAP signing, if requested by Active Directory. CVS-SW does not support LDAP signing.

> unixUserPassword is queried by LDAP and is not sent in plaintext but instead in a salted hash. By default, Windows LDAP does not populate the unixUserPassword fields. This field is only required if you need to leverage Windows LDAP for interactive logins through LDAP to clients. Cloud Volumes Service does not support interactive LDAP logins to the instances.

The following figure shows a packet capture from an NFS Kerberos conversation next to a capture of NFS over AUTH_SYS. Note how the information available in a trace differs between the two and how enabling in-flight encryption offers greater overall security for NAS traffic.

IP addresses of the NFS client and CVS instance

Genericized NFS call/reply

No.	Time	Source	Destination	Protocol	Length	Info
380	9.218014	10.193.67.225	10.193.67.219	NFS	346	V4 Call (Reply In 381)
381	9.218480	10.193.67.219	10.193.67.225	NFS	426	V4 Reply (Call In 380)
382	9.218641	10.193.67.225	10.193.67.219	NFS	370	V4 Call (Reply In 397)
397	9.369035	10.193.67.219	10.193.67.225	NFS	458	V4 Reply (Call In 382)

```
> Frame 381: 426 bytes on wire (3408 bits), 426 bytes captured (3408 bits)
> Ethernet II, Src: IntelCor_7f:da:bc (90:e2:ba:7f:da:bc), Dst: VMware_a0:2c:2d (00:50:56:a0:2c:2d)
> Internet Protocol Version 4, Src: 10.193.67.219, Dst: 10.193.67.225
> Transmission Control Protocol, Src Port: 2049, Dst Port: 738, Seq: 6305, Ack: 6569, Len: 360
> Remote Procedure Call, Type:Reply XID:0xef5e998d
v GSS-Wrap
    Length: 300
    GSS Data: 050407ff0000000000000000025913451ee1d43d298cf3031…
  > krb5_blob: 050407ff0000000000000000025913451ee1d43d298cf3031…
v Network File System
    [Program Version: 4]
    [V4 Procedure: COMPOUND (1)]
```

GSS wrapped NFS calls/replies with no other identifying information

IP addresses of the NFS client and CVS instance

Detailed NFS call types and file handle information

No.	Time	Source	Destination	Protocol	Length	Info
33	0.958480	10.193.67.201	10.193.67.204	NFS	458	V4 Reply (Call In 32) OPEN StateID: 0x0481
34	0.958784	10.193.67.204	10.193.67.201	NFS	306	V4 Call (Reply In 35) SETATTR FH: 0x6c07918a
35	0.959284	10.193.67.201	10.193.67.204	NFS	358	V4 Reply (Call In 34) SETATTR

```
> Opcode: PUTFH (22)
> Opcode: SETATTR (34)
v Opcode: GETATTR (9)
    Status: NFS4_OK (0)
  v Attr mask[0]: 0x0010011a (Type, Change, Size, FSID, FileId)
    > reqd_attr: Type (1)
    > reqd_attr: Change (3)
    > reqd_attr: Size (4)
    > reqd_attr: FSID (8)
    v reco_attr: FileId (20)
        fileid: 9232254136597092620
  v Attr mask[1]: 0x00b0a03a (Mode, NumLinks, Owner, Owner_Group, Space_Used, Time_Access, Time_Metadata, Time_Modify, Mounted_on_FileId)
    v reco_attr: Mode (33)
      > mode: 0644, Name: Unknown, Read permission for owner, Write permission for owner, Read permission for group, Read permission for others
    > reco_attr: NumLinks (35)
    v reco_attr: Owner (36)
      > fattr4_owner: root@NTAP.LOCAL
    v reco_attr: Owner_Group (37)
      > fattr4_owner_group: root@NTAP.LOCAL
    > reco_attr: Space_Used (45)
    > reco_attr: Time_Access (47)
    > reco_attr: Time_Metadata (52)
    > reco_attr: Time_Modify (53)
    > reco_attr: Mounted_on_FileId (55)
```

File ID

Permission information

Owner and group ID strings

VM network interfaces

One trick attackers might attempt is to add a new network interface card (NIC) to a VM in promiscuous mode (port mirroring) or enable promiscuous mode on an existing NIC in order to sniff all traffic. In Google Cloud, adding a new NIC requires a VM to be shut down entirely, which creates alerts, so attackers cannot do this

unnoticed.

In addition, NICs cannot be set to promiscuous mode at all and will trigger alerts in Google Cloud.

Control plane architecture

All management actions to Cloud Volumes Service are done through API. Cloud Volumes Service management integrated into the GCP Cloud Console also uses the Cloud Volumes Service API.

Identity and Access Management

Identity and Access Management (IAM) is a standard service that enables you to control authentication (logins) and authorization (permissions) to Google Cloud project instances. Google IAM provides a full audit trail of permissions authorization and removal. Currently Cloud Volumes Service does not provide control plane auditing.

Authorization/permission overview

IAM offers built-in, granular permissions for Cloud Volumes Service. You can find a complete list of granular permissions here.

IAM also offers two predefined roles called `netappcloudvolumes.admin` and `netappcloudvolumes.viewer`. These roles can be assigned to specific users or service accounts.

Assign appropriate roles and permission to allow IAM users to manage Cloud Volumes Service.

Examples for using granular permissions include the following:

- Build a custom role with only get/list/create/update permissions so that users cannot delete volumes.
- Use a custom role with only `snapshot.*` permissions to create a service account that is used to build application- consistent Snapshot integration.
- Build a custom role to delegate `volumereplication.*` to specific users.

Service accounts

To make Cloud Volumes Service API calls through scripts or Terraform, you must create a service account with the `roles/netappcloudvolumes.admin` role. You can use this service account to generate the JWT tokens required to authenticate Cloud Volumes Service API requests in two different ways:

- Generate a JSON key and use Google APIs to derive a JWT token from it. This is the simplest approach, but it involves manual secrets (the JSON key) management.
- Use Service account impersonation with `roles/iam.serviceAccountTokenCreator`. The code (script, Terraform, and so on.) runs with Application Default Credentials and impersonates the service account to gain its permissions. This approach reflects Google security best practices.

See Creating your service account and private key in the Google cloud documentation for more information.

Cloud Volumes Service API

Cloud Volumes Service API uses a REST-based API by using HTTPS (TLSv1.2) as the underlying network

transport. You can find the latest API definition here and information about how to use the API at Cloud Volumes APIs in the Google cloud documentation.

The API endpoint is operated and secured by NetApp using standard HTTPS (TLSv1.2) functionality.

JWT tokens

Authentication to the API is performed with JWT bearer tokens (RFC-7519). Valid JWT tokens must be obtained by using Google Cloud IAM authentication. This must be done by fetching a token from IAM by providing a service account JSON key.

Audit logging

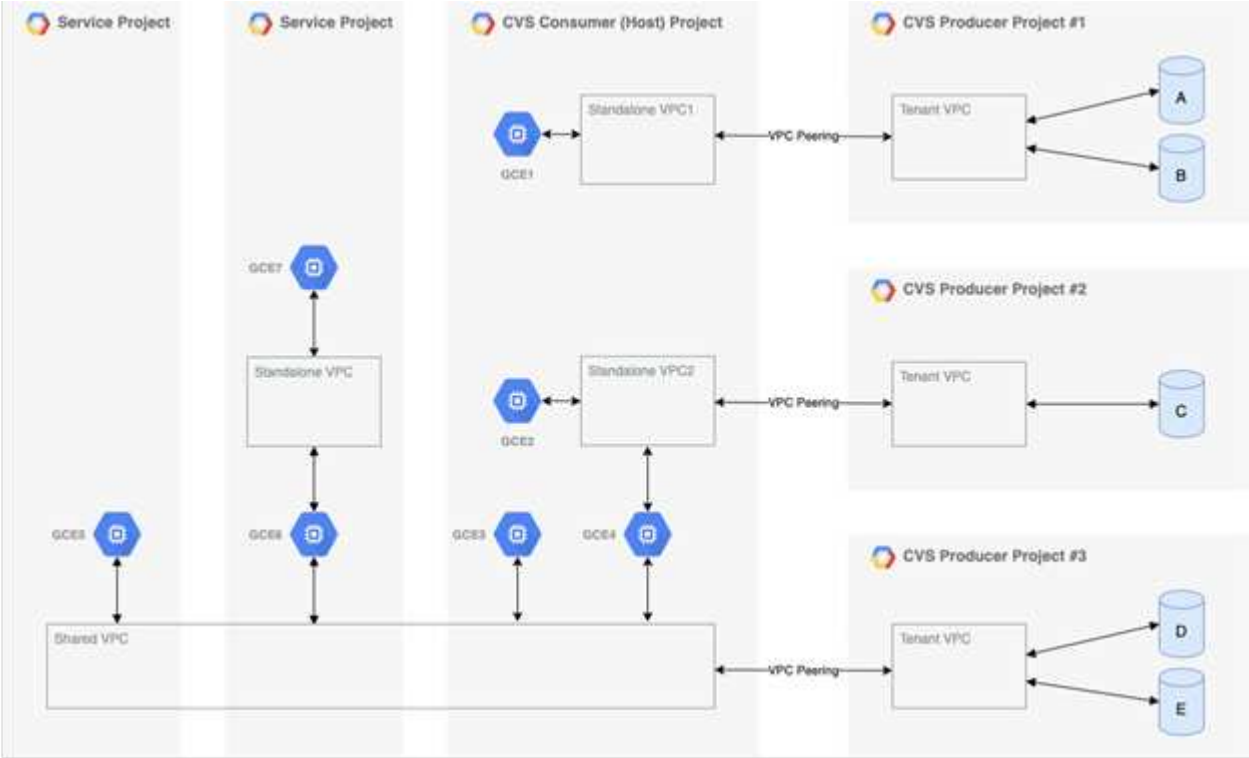
Currently, no user-accessible control plane audit logs are available.

Data plane architecture

Cloud Volumes Service for Google Cloud leverages the Google Cloud private services access framework. In this framework, users can connect to the Cloud Volumes Service. This framework uses Service Networking and VPC peering constructs like other Google Cloud services, ensuring complete isolation between tenants.

For an architecture overview of Cloud Volumes Service for Google Cloud, see Architecture for Cloud Volumes Service.

User VPCs (standalone or shared) are peered to VPCs within Cloud Volumes Service managed tenant projects, which hosts the volumes.



The preceding figure shows a project (the CVS consumer project in the middle) with three VPC networks connected to Cloud Volumes Service and multiple Compute Engine VMs (GCE1-7) sharing volumes:

- VPC1 allows GCE1 to access volumes A and B.
- VPC2 allows GCE2 and GCE4 to access volume C.
- The third VPC network is a shared VPC, shared with two service projects. It allows GCE3, GCE4, GCE5, and GCE6 to access volumes D and E. Shared VPC networks are only supported for volumes of the CVS-Performance service type.

> GCE7 cannot access any volume.

Data can be encrypted both in-transit (using Kerberos and/or SMB encryption) and at-rest in Cloud Volumes Service.

Data encryption in transit

Data in transit can be encrypted at the NAS protocol layer, and the Google Cloud network itself is encrypted, as described in the following sections.

Google Cloud network

Google Cloud encrypts traffic on the network level as described in Encryption in transit in the Google documentation. As mentioned in the section “Cloud Volumes Services architecture,” Cloud Volumes Service is delivered out of a NetApp-controlled PSA producer project.

In case of CVS-SW, the producer tenant runs Google VMs to provide the service. Traffic between user VMs and Cloud Volumes Service VMs is encrypted automatically by Google.

Although the data path for CVS-Performance isn’t fully encrypted on the network layer, NetApp and Google use a combination of IEEE 802.1AE encryption (MACSec), encapsulation (data encryption), and physically restricted networks to protect data in transit between the Cloud Volumes Service CVS-Performance service type and Google Cloud.

NAS protocols

NFS and SMB NAS protocols provide optional transport encryption at the protocol layer.

SMB encryption

SMB encryption provides end-to-end encryption of SMB data and protects data from eavesdropping occurrences on untrusted networks. You can enable encryption for both the client/server data connection (only available to SMB3.x capable clients) and the server/domain controller authentication.

When SMB encryption is enabled, clients that do not support encryption cannot access the share.

Cloud Volumes Service supports RC4-HMAC, AES-128-CTS-HMAC-SHA1, and AES-256-CTS-HMAC-SHA1 security ciphers for SMB encryption. SMB negotiates to the highest supported encryption type by the server.

NFSv4.1 Kerberos

For NFSv4.1, CVS-Performance offers Kerberos authentication as described in RFC7530. You can enable Kerberos on a per-volume basis.

The current strongest available encryption type for Kerberos is AES-256-CTS-HMAC-SHA1. NetApp Cloud Volumes Service supports AES-256-CTS-HMAC-SHA1, AES-128-CTS-HMAC-SHA1, DES3, and DES for

NFS. It also supports ARCFOUR-HMAC (RC4) for CIFS/SMB traffic, but not for NFS.

Kerberos provides three different security levels for NFS mounts that offer choices for how strong the Kerberos security should be.

As per RedHat's Common Mount Options documentation:

```
sec=krb5 uses Kerberos V5 instead of local UNIX UIDs and GIDs to
authenticate users.
sec=krb5i uses Kerberos V5 for user authentication and performs integrity
checking of NFS operations using secure checksums to prevent data
tampering.
sec=krb5p uses Kerberos V5 for user authentication, integrity checking,
and encrypts NFS traffic to prevent traffic sniffing. This is the most
secure setting, but it also involves the most performance overhead.
```

As a general rule, the more the Kerberos security level has to do, the worse the performance is, as the client and server spend time encrypting and decrypting NFS operations for each packet sent. Many clients and NFS servers provide support for AES-NI offloading to the CPUs for a better overall experience, but the performance impact of Kerberos 5p (full end-to-end encryption) is significantly greater than the impact of Kerberos 5 (user authentication).

The following table shows differences in what each level does for security and performance.

Security level	Security	Performance
NFSv3—sys	• Least secure; plain text with numeric user IDs/group IDs • Able to view UID, GID, client IP addresses, export paths, file names, permissions in packet captures	• Best for most cases
NFSv4.x—sys	• More secure than NFSv3 (client IDs, name string/domain string matching) but still plain text • Able to view UID, GID, client IP addresses, name strings, domain IDs, export paths, file names, permissions in packet captures	• Good for sequential workloads (such as VMs, databases, large files) • Bad with high file count/high metadata (30-50% worse)

Security level	Security	Performance
NFS—krb5	• Kerberos encryption for credentials in every NFS packet—wraps UID/GID of users/groups in RPC calls in GSS wrapper • User requesting access to mount needs a valid Kerberos ticket (either through username/password or manual key tab exchange); ticket expires after a specified time period and user must reauthenticate for access • No encryption for NFS operations or ancillary protocols like mount/portmapper/nlm (can see export paths, IP addresses, file handles, permissions, file names, atime/mtime in packet captures)	• Best in most cases for Kerberos; worse than AUTH_SYS
NFS—krb5i	• Kerberos encryption for credentials in every NFS packet—wraps UID/GID of users/groups in RPC calls in GSS wrapper • User requesting access to mount needs a valid Kerberos ticket (either via username/password or manual key tab exchange); ticket expires after a specified time period and user must reauthenticate for access • No encryption for NFS operations or ancillary protocols like mount/portmapper/nlm (can see export paths, IP addresses, file handles, permissions, file names, atime/mtime in packet captures) • Kerberos GSS checksum is added to every packet to ensure nothing intercepts the packets. If checksums match, conversation is allowed.	• Better than krb5p because the NFS payload is not encrypted; only added overhead compared to krb5 is the integrity checksum. Performance of krb5i won’t be much worse than krb5 but will see some degradation.

Security level	Security	Performance
NFS – krb5p	• Kerberos encryption for credentials in every NFS packet—wraps UID/GID of users/groups in RPC calls in GSS wrapper • User requesting access to mount needs a valid Kerberos ticket (either via username/password or manual keytab exchange); ticket expires after specified time period and user must reauthenticate for access • All of the NFS packet payloads are encrypted with the GSS wrapper (cannot see file handles, permissions, file names, atime/mtime in packet captures). • Includes integrity check. • NFS operation type is visible (FSINFO, ACCESS, GETATTR, and so on). • Ancillary protocols (mount, portmap, nlm, and so on) are not encrypted - (can see export paths, IP addresses)	• Worst performance of the security levels; krb5p has to encrypt/decrypt more. • Better performance than krb5p with NFSv4.x for high file count workloads.

In Cloud Volumes Service, a configured Active Directory server is used as Kerberos server and LDAP server (to lookup user identities from an RFC2307 compatible schema). No other Kerberos or LDAP servers are supported. NetApp highly recommends that you use LDAP for identity management in Cloud Volumes Service. For information on how NFS Kerberos is shown in packet captures, see the section “Packet sniffing/trace considerations.”

Data encryption at rest

All volumes in Cloud Volumes Service are encrypted-at-rest using AES-256 encryption, which means all user data written to media is encrypted and can only be decrypted with a per-volume key.

- For CVS-SW, Google-generated keys are used.
- For CVS-Performance, the per-volume keys are stored in a key manager built into the Cloud Volumes Service.

Starting in November 2021, preview customer-managed encryption keys (CMEK) functionality was made available. This enables you to encrypt the per-volume keys with a per-project, per-region master key that is hosted in Google Key Management Service (KMS). KMS enables you to attach external key managers.

For information about configuring KMS for CVS-Performance, see Setting up customer-managed encryption keys.

Firewall

Cloud Volumes Service exposes multiple TCP ports to serve NFS and SMB shares:

- Ports required for NFS access
- Ports required for SMB access

Additionally, SMB, NFS with LDAP including Kerberos, and dual-protocol configurations require access to a Windows Active Directory domain. Active Directory connections must be configured on a per-region basis. Active Directory Domain controllers (DC) are identified by using DNS-based DC discovery using the specified DNS servers. Any of the DCs returned are used. The list of eligible DCs can be limited by specifying an Active Directory site.

Cloud Volumes Service reaches out with IP addresses from the CIDR range allocated with the `gcloud compute address` command while on-boarding the Cloud Volumes Service. You can use this CIDR as source addresses to configure inbound firewalls to your Active Directory domain controllers.

Active Directory Domain Controllers must expose ports to the Cloud Volumes Service CIDRs as mentioned here.

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